Alex Dils

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Education

University of California, Berkeley May 2028
B.A. in ***Computer Science*** | GPA: 3.7/4.0
Coursework: Data Structures & Algorithms, Machine Structures, Discrete Mathematics & Probability, Multivariable Calculus
Activities: Machine Learning at Berkeley (ML@B), Internal & Research Committees

Experience

Human Archive
Oct 2026 – Present
Robotics Research – Intern
- Benchmarked **SLAM and 3D hand-tracking pipelines** on 30/60 fps egocentric video across cloud GPUs, measuring throughput and cost per processed video-hour.
- Tuned CPU/GPU allocation and processing concurrency; compared outputs against H100 baselines to assess reconstruction consistency and recommend cost-efficient configurations.
- Evaluated models and ablated training data sources, pretraining strategies, and fine-tuning methods.

Stanford Division of Computational Medicine
Jun 2022 – Present
Computer Vision Research – Intern
- Developed domain-robust skin-lesion segmentation methods and finite-element lung-motion augmentations that **improved SoTA tumor-segmentation Dice by >0.05**.
- Research featured on Stanford Medicine homepage
- Built a deployment tool for a model that generates corresponding spot-compression mammograms from CC and MLO views, making it deployable to clinicians and patients.

Stanford Canary Center / Gevaert Lab
Jun 2026 – Aug 2026
Computer Vision Research – Canary CREST Research Scholar
- Developed and deployed volume estimation on cNF Vision, a clinical tool deployed to 50+ clinicians and patients.
- Improved lesion-volume estimation R^2 **from 0.78 to 0.85** on 40 held-out real images by building a physics-guided generative AI pipeline producing **42,305 synthetic 3D lesions** and **279 longitudinal sequences**.
- Reduced average depth-estimation RMSE by **72.9%** (47.30 to 12.83 cm) on scans with simulated lesions by fine-tuning four depth models on synthetic 3D cNF data.
- Built a clinical tool for viewing and editing PET and CT scan annotations using JavaScript and Electron.

Logitech
Feb 2026 – Jun 2026
Applied Machine Learning – Machine Learning Engineer (Contract)
- Built a **VLM evaluation harness** for meeting-room readiness and participant classification on Logitech's Rally Board 65, measuring task accuracy and inference latency across 55 models.
- Ran model and prompt ablations and identified Qwen3.5 4B as the leading deployment candidate for the target hardware and task.

Technical Skills

Tools: AI agents, orchestration, multi-agent workflows, Codex, Claude, Docker, Slurm, Git, Vercel, Electron
Languages: Python, C, JavaScript, TypeScript; **ML/CV:** PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, SimpleITK

Projects

Openleaf | alex-dils.com/openleaf 2026
- Build and maintain an opensource macOS research workspace unifying LaTeX, PDF, Python, GitHub, SSH, and coding agents.

LanternTrace Explorer | alex-dils.com/lanterntrace 2026
- Built a physics-based explorer to track and forecast spotted lanternfly spread across the U.S. East Coast.

Publications

★ *indicates first authorship.*

- ★ **Synthesizing Longitudinal Cutaneous Neurofibroma Imaging for Digital-Twin Burden Quantification**. MICCAI MWM 2026.
  Christoph Sadée, **Alex Dils**, Krish Sangani, Lillian Rubino, Alexander J Ferenchick, Varun Wadhwa, et al.
- ★ **Microplastic Identification Using AI-Driven Image Segmentation and GAN-Generated Ecological Context**. arXiv preprint, 2024.
  Alex Dils, David Raymond, Jack Spottiswood, Samay Kodige, Dylan Karmin, Rikhil Kokal, Win Cowger, and Chris Sadée.
- ★ **Eye For An Eye: A Deep-Learning and Analytical Method to Spatializing Stereoscopic Images**. National High School Journal of Science, 2025.
  Jake Yoshinaka, **Alex Dils**, and Leo McDonnell.
- **Addressing Bias and Confounders in AI-based Image Diagnosis — A Study of 117,610 Skin Lesions**. Manuscript under revision at *Cell*, 2025.
  Christoph Sadée, Serena Zhang, **Alex Dils**, Jared Bitz, Stefano Testa, Zhuo Ran Cai, et al.
- **Integrating Mechanistic Knowledge into Deep Learning for Improved Cancer Detection**. FEniCS Conference Proceedings, 2024.
  Christoph Sadée, Katherine Hartmann, **Alex Dils**, Ianto Xi, Francisco Carrillo-Perez, et al.